FARMSKING TECHNOLOGIES PRIVATE LIMITED

Master Project Working Tree • Developer Operational Runtime Blueprint

CIN: U01100PB2026PTC063294
A4 Standard: 210mm × 297mm | ISO 9001 / 27001
Target Audience: Core Engineering & Platform Developers

1. System Runtime Execution Flow (How a Single Request Operates)

Client → API Gateway → Database → Audit

```
[ CLIENT LAYER (Expo Web / Mobile) ]
  │ Farmer / Buyer / Partner / Admin Browser (Port: 8081 / 8082)
  │
  ├── 1. UI Screen (e.g., memberships.tsx, ai-doctor.tsx, checkout.tsx)
  ├── 2. React Hook (e.g., useFarmerPlans(), useOrders(), useWallet())
  └── 3. Axios API Client (Injects Bearer JWT, handles 10s timeouts, formats JSON)
        │
        ▼ HTTP REST POST / GET / PATCH (Port: 3000)
[ API GATEWAY LAYER (NestJS Enterprise Framework) ]
  ├── 4. CORS & Security Headers (Authorizes frontend/admin domains)
  ├── 5. Global ValidationPipe (Validates DTOs, strips unauthorized payload fields)
  ├── 6. JwtAuthGuard & RolesGuard (Enforces RBAC: SUPERADMIN, FARMER, SELLER, etc.)
  ├── 7. AuditLoggingInterceptor (Logs actor King ID, IP, table name, action & timestamp)
  ├── 8. Domain Controller (Maps endpoint e.g., @Get('upgrade-quote'), @Post('initiate'))
  └── 9. Domain Service (Executes business logic, prorated math, 1% TCS, 18% GST, escrow)
        │
        ▼ Prisma Type-Safe Query Engine
[ DATA PERSISTENCE LAYER (PostgreSQL - 103 Tables) ]
  ├── Master Entities (User, FarmerPlan, Order, SubOrder, CropCycle, WalletTransaction)
  ├── Financial Escrow & Payouts (EscrowHold, Gstr8Report, SellerPayout, BankBeneficiary)
  └── System Audit Trail (AuditLog table recording every single create/update/delete)
        │
        ▼ JSON Response (HTTP 200/201)
[ React UI Rerenders with Live State ]
```

2. Master User Operational Lifecycle Tree (9-Stage Identity Flow)

Single Identity Multi-Role Progression

Stage 1 & 2 **Guest Visitor → Registered Customer (King ID Generated)** — **Role: CUSTOMER**

Visitor lands on platform • Registers via Mobile Number & OTP • System assigns permanent King ID (FK-YYYY-XXXXX) • Can browse AgriStore, add seeds/fertilizers to cart, and checkout with UTR payment submission.

Stage 3 **Verified Farmer Onboarding & Agronomy Records** — **Role: FARMER**

Farmer enters landholdings, acreage, soil chemistry, and irrigation types • System unlocks ICAR-aligned Crop Calendar • Farmer logs sowing, expenses, labour, spray, and fertilizer for active Crop Cycles.

Stage 4 **AI Crop Doctor Diagnostic & Agronomy Advisory** — **AI Doctor / Advisor**

Farmer snaps leaf photo • AI neural vision diagnoses disease with confidence score • Assigned Agronomy Advisor reviews and issues digital prescription • Farmer can request Doctor Switch in 1 click.

Stage 5 **Farmer VIP Membership Upgrade (Prorated Credits & Downgrade Lock)** — **Plan Engine**

Farmer upgrades from FREE to SILVER/GOLD/PLATINUM • Backend calculates prorated credit of unused days from previous plan and deducts from upgrade total • Downgrade is strictly prohibited and locked.

Stage 6 & 7 **Farmer Becomes Seller → Multi-Vendor E-Commerce Payouts** **Role: SELLER**

Farmer submits GSTIN/PAN and Bank KYC • Admin approves vendor storefront in admin-sellers.tsx • Seller publishes products or wholesale lots • Orders split into vendor SubOrder items • Funds held in Escrow • Upon OTP delivery, system automatically deducts 1% statutory TCS (Sec 194-O) + platform fee, crediting net amount to Seller Wallet for IMPS/NEFT payout.

Stage 8 & 9 **Referral Royalty Engine → King Connect B2B Mandi Shared Ledger** **Viral & B2B**

Farmer shares referral code • Purchases by referred network instantly trigger royalty deposits into referrer's digital wallet • On King Connect B2B Mandi, farmer lists bulk harvest (e.g. 500 quintals) • Institutional traders submit Demand Requests • Double-entry P2P ledger synchronizes contracts between buyer and seller.

3. Feature Operational Working Trees (For Developers)

Concrete Execution Paths

Workflow A: AI Crop Doctor Cycle

```
[ Farmer snaps leaf photo ]
  |
  ▼
frontend/ai-doctor.tsx
  | Calls diagnoseCropApi(formData)
  ▼
AiDoctorService.diagnose
  | Neural vision detects pathogen
  | Computes Confidence Score (94%)
  ▼
AiCropDiagnostic Table
  | Inserts scan record + symptoms
  ▼
Doctor Review Queue
  | Doctor issues chemical prescription
  ▼
Push Notification to Farmer
```

Workflow B: AgriStore Split-Orders

```
[ Buyer Cart with Multiple Vendors ]
  |
  ▼
agristore/checkout.tsx
  | Buyer submits UTR transaction ref
  ▼
OrdersService.createMasterOrder
  | Master Order: PENDING_VERIFICATION
  | Splits into vendor SubOrder records
  | Funds locked in EscrowHold table
  ▼
Admin verifies UTR in admin-orders
  | Moves Order status to PROCESSING
  ▼
Seller Dispatches • Delivery OTP
  | Deducts 1% TCS + 5% Fee
  | Net credited to Seller Wallet
```

Workflow C: VIP Plan Upgrade Engine

```
[ Farmer requests Plan Upgrade ]
  |
  ▼
GET /farmer-plans/upgrade-quote
  | Unused days = Total - Elapsed
  | Daily rate = Previous Plan Price / 365
  | Prorated credit = Unused days × Rate
  | Net Price = Target Price - Credit
  ▼
Downgrade Guard Enforcement
  | Target level < Current level?
  | → THROW 400 Bad Request ("Locked")
  ▼
POST /farmer-plans/apply-change
  | User.farmerTier updated
  | Old plan moved to FarmerPlanHistory
```

Workflow D: Advisor Switch Queue

```
[ Farmer requests Doctor Switch ]
  | Submits reason & preferred advisor
  ▼
POST /doctor-changes/request
  | Inserts into DoctorChangeRequest
  | Status: PENDING
  ▼
Admin Console: admin/memberships
  | Pending Doctor Switch Requests queue
  | Shows Farmer, Old & New Doctor
  ▼
1-Click Approve / Decline Action
  | Updates User.assignedDoctorId
  | Sends push alerts to all parties
```

4. Mandatory System Audit Logging Tree (All 103 Tables)

AuditLoggingInterceptor → AuditLog Table → Admin Console

```
[ ANY Mutation Across 103 Database Tables ] (CREATE, UPDATE, DELETE, STATUS_CHANGE)
  │
  ▼
AuditLoggingInterceptor (backend/src/common/interceptors/audit-logging.interceptor.ts)
  │ Extracts:
  │   • Actor King ID (User / Admin making modification)
  │   • Table Name (e.g., User, FarmerPlan, CropCycle, Order, WalletTransaction)
  │   • Action Type (CREATE / UPDATE / DELETE / STATUS_CHANGE)
  │   • Client IP Address & User Agent
  │   • Current Date & Time (ISO 8601 Timestamp)
  │   • Record Diff (State before change vs State after change)
  │
  ▼
PostgreSQL AuditLog Table (backend/prisma/schema/section_860.prisma)
  │ Writes permanent immutable audit record
  │ Indexed by: tableName, actorId, createdAt
  │ Guarantees zero unmonitored financial or agronomy data changes
  │
  ▼
Admin Console: admin/new-settings.tsx • [System Logs] Tab
  │ Rendered by SettingAuditLogsView.tsx
  │ Admin can filter and search across all 103 tables in real-time
  │ Exportable A4 audit trail for legal and ISO compliance
```

5. Developer Golden Rules & Operational Invariants

Master Verification Checklist

Rule 1: 100% Pure English UI & Simple Wording

Strict Language Rule

All user-facing UI copy, buttons, badges, diagnostic messages, and alerts must be in **100% pure professional English**. Do not hardcode vernacular text in core components. The only component allowed to display regional scripts is the dedicated LanguageSelector box.

Rule 2: Primary Laptop/PC View Priority

Responsive Benchmark

The user tests on desktop and laptop screens. All layouts, headers, cards, and data tables must be visually balanced and unclipped on desktop view first. Mobile responsiveness is maintained automatically in the background.

Rule 4: Strict A4 Page Dimensions (210mm × 297mm)

Document & PDF Standard

Every printable report, invoice, diagnostic summary, or architecture dossier must specify @page { size: A4; margin: 12mm 15mm; } with page-break-inside: avoid to guarantee standard A4 delivery.

Rule 6: Zero Automatic Live Production Deployment

Local Verification First

Never run automated live deploys (git push hostinger) during routine feature development. Always verify locally on localhost:3000 (API), localhost:8081 (Frontend), and localhost:8082 (Admin). Deploy only when explicitly instructed by the user.